

Data Collection and Preprocessing Phase

Date	27 June 2025
Student ID	SWUID20250181505
Project Title	online payments fraud detection using machine learning
Maximum Marks	2 Marks

Data Collection Plan & Raw Data Sources Identification Template

Elevate your data strategy with the Data Collection plan and the Raw Data Sources report, ensuring meticulous data curation and integrity for informed decision-making in every analysis and decision-making endeavor.

Data Collection Plan Template

Section	Description
Project Overview	The Online Payments Fraud Detection System is a machine learning project aimed at identifying fraudulent online payment transactions in real-time using a Random Forest Classifier. The system integrates a Flask-based web application to allow users (e.g., small business owners, financial institutions) to input transaction details and receive instant fraud predictions. The primary objective is to enhance financial security by accurately detecting fraud while providing a user-friendly interface for transaction verification.

Data Collection Plan	The primary data source is a synthetic dataset from Kaggle, derived from the PaySim simulator, which mimics mobile money transactions based on real financial logs from a mobile money service in an African country.
Raw Data Sources Identified	Financial Fraud Detection Dataset

Raw Data Sources Template

Source Name	Description	Location/URL	Format	Size	Access Permissions
Financial Fraud Detection Dataset	A synthetic dataset from Kaggle, generated by the PaySim simulator, containing 1,048,575 mobile money transaction records over 30 days.	https://www.kaggle.com/datasets/sriharshaedala/financial-fraud-detection-dataset	CSV	493.53 MB	Public